MIS 311 – ASSIGNMENT #1
STUDENT PERFORMANCE DATA ANALYSIS

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1. Data Overview

The Student Performance dataset contains 199 student records and 8 columns in the current cleaned dataset. The variables include gender, race/ethnicity, parental level of education, mathematics score, reading score, writing score, total score, and average score. The dataset can be used to summarize academic performance and compare average scores across race/ethnicity groups.

2. Data Cleaning

The original dataset was inspected for missing values and duplicate records. Duplicate records were checked using Excel's Remove Duplicates function, with all 8 columns selected. The cleaned dataset contains 199 records. Missing values in parental_level_of_education should be labelled "Unknown" where the information was not provided. Missing average_score values can be recalculated from the mathematics, reading, and writing scores when all three subject scores are available. The cleaned data should be checked again to confirm that missing values and duplicates have been handled.

3. Descriptive Statistics

The descriptive statistics show that reading had the highest mean score (67.63), followed by writing (66.37) and mathematics (64.22). The mean scores are based on 199 records for each subject. Mathematics also had the largest standard deviation among the three subjects (approximately 15.90), while reading's standard deviation was approximately 15.78 and writing's was approximately 16.26. These values indicate variation in student scores across all three subjects.

4. Key Insights

Insight 1: Average Scores by Subject

The average scores were compared by calculating the mean for mathematics, reading, and writing. Reading had the highest average score at 67.63, while mathematics had the lowest at 64.22, a difference of about 3.41 points. This suggests that students scored higher on average in reading than in mathematics in this dataset. However, the comparison alone does not establish why this difference occurred.

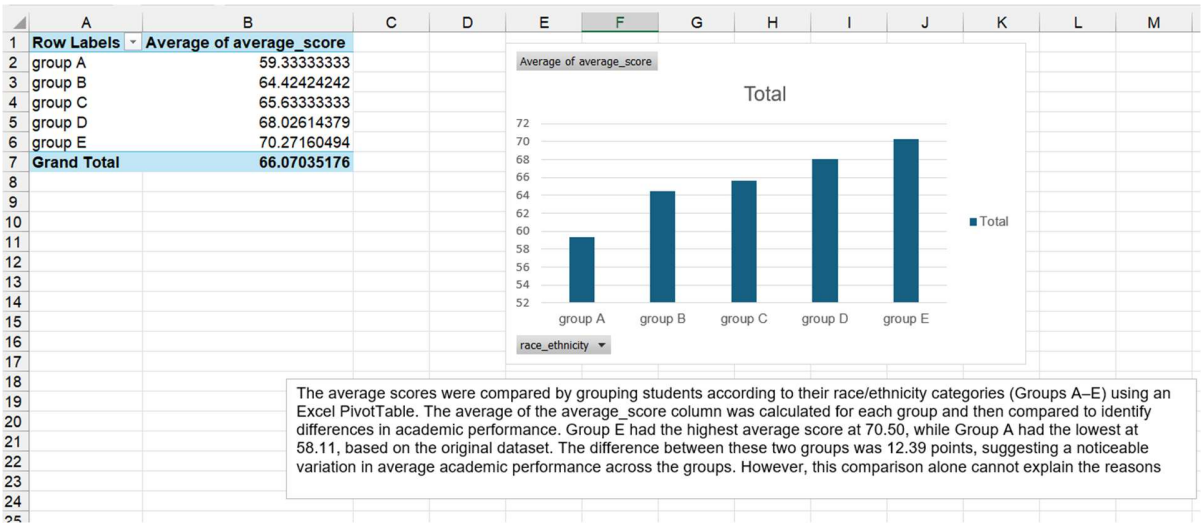
Insight 2: Average Scores by Race/Ethnicity Group

An Excel PivotTable was used to group students by race/ethnicity and calculate the average of average_score for each group. Group E had the highest average (70.27), while Group A had the lowest (59.33), a difference of about 10.94 points. Groups B, C, and D had average scores of approximately 64.42, 65.63, and 68.03, respectively. This comparison shows differences in average scores between the groups, but it does not explain the causes of those differences.

5. Visualizations

Figure 1. Average Score by Race/Ethnicity Group

The column chart compares the average student score across Groups A–E. The bars increase from Group A to Group E, with Group E showing the highest average in the PivotTable.



Row Labels	Average of average_score
group A	59.33333333
group B	64.42424242
group C	65.63333333
group D	68.02614379
group E	70.27160494
Grand Total	**66.07035176**

The average scores were compared by grouping students according to their race/ethnicity categories (Groups A–E) using an Excel PivotTable. The average of the average_score column was calculated for each group and then compared to identify differences in academic performance. Group E had the highest average score at 70.50, while Group A had the lowest at 58.11, based on the original dataset. The difference between these two groups was 12.39 points, suggesting a noticeable variation in average academic performance across the groups. However, this comparison alone cannot explain the reasons

Figure 2. Distribution of Students' Average Scores

The distribution chart groups average scores into six ranges. The largest number of students falls in the 60–69 range (48 students), followed by 70–79 (46 students) and 50–59 (38 students). The 90–100 range contains the fewest students (10). The counts across the six ranges total 199 students.

Score Range	Distribution of Students' Average Scores
0–49	29
50–59	38
60–69	48
70–79	46
80–89	28
90–100	10

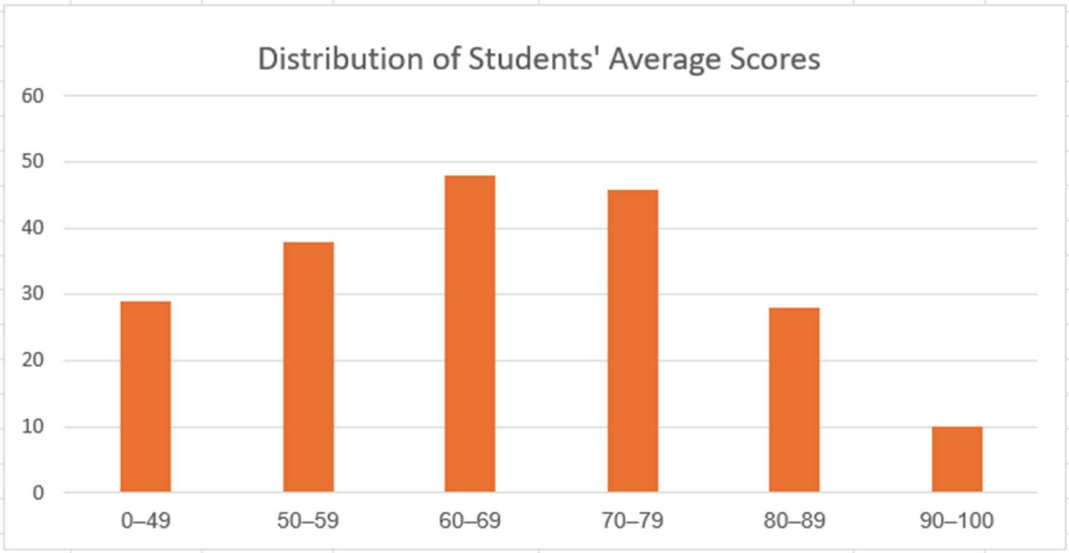


Figure 3. Descriptive Statistics for Mathematics, Reading, and Writing

The table below summarizes the descriptive statistics calculated in Excel for the three subject-score variables.

math_score		*reading_score*		*writing_score*	
Mean	64.2160804	Mean	67.6281407	Mean	66.36683417
Standard Error	1.127361218	Standard Error	1.118896059	Standard Error	1.152591036
Median	65	Median	68	Median	68
Mode	58	Mode	67	Mode	74
Standard Deviation	15.90338706	Standard Deviation	15.78397129	Standard Deviation	16.25929743
Sample Variance	252.9177199	Sample Variance	249.1337496	Sample Variance	264.3647531
Kurtosis	0.946460599	Kurtosis	-0.140521623	Kurtosis	0.072613787
Skewness	-0.366830333	Skewness	-0.231437425	Skewness	-0.33337789
Range	100	Range	83	Range	90
Minimum	0	Minimum	17	Minimum	10
Maximum	100	Maximum	100	Maximum	100
Sum	12779	Sum	13458	Sum	13207
Count	199	Count	199	Count	199

The descriptive statistics reveal that Reading achieved the highest average score of 67.47, followed by Writing at 66.18 and Mathematics at 64.18. The 3.29-point difference between Reading and Mathematics suggests that students performed relatively better in Reading than in Mathematics in the original dataset. This finding highlights the importance of identifying potential learning difficulties in Mathematics and developing targeted support strategies to improve students' academic performance.

6. Conclusion

Overall, the analysis indicates that reading had the highest mean score and mathematics had the lowest mean score among the three subjects. The group comparison also shows that Group E had the highest average_score and Group A the lowest in the current PivotTable. The score distribution is concentrated mainly between 60 and 79. These findings summarize patterns in the dataset, but further analysis would be needed to investigate the factors associated with the differences.

Before submission, confirm that all missing values have been handled in the final cleaned workbook, add your name and IRN, and include the GitHub portfolio link required by the assignment.